Varun Belagali

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Education

Stony Brook University, NY, USA **Aug 2022 – Present**
Doctor of Philosophy, Computer Science
Advised by Prof. Dimitris Samaras in Computer Vision Lab
Research Interest: **Visual representation learning with self-supervision**
Coursework: Machine Learning, Computer Vision, Robotics, Distributed Systems, Database Systems (GPA 4/4)

R V College of Engineering, India **Aug 2016 – Aug 2020**
Bachelor of Engineering, Computer Science and Engineering
Coursework: Operating Systems, Analysis of Algorithms, Neural Networks, Data Structures, Compilers (GPA 9.22/10)

Research Papers (selected)

1. **CDG-MAE: Learning Correspondences from Diffusion Generated Views** [Link] arXiv 2025
   Varun Belagali, P. Marza, S. Yellapragada, Z. Li, T. Nandi, R. Madduri, J. Saltz, S. Christodoulidis, M. Vakalopoulou, D. Samaras. NeurIPS reject with scores (4,4,4,5/6)
2. **Gen-SIS: Generative Self-augmentation Improves Self-supervised Learning** [Link] arXiv 2024
   Varun Belagali*, S. Yellapragada*, A. Graikos, S. Kapse, Z. Li, T. Nandi, R. Madduri, P. Prasanna, J. Saltz, D. Samaras. NeurIPS reject with scores (3,4,4,4/6)
3. **HyperMAE: Modulating Implicit Neural Representations for MAE Training** [Link] 2023
   Varun Belagali, L. Zhou, X. Li,, D. Samaras. **NeurIPS Workshop -** Self-supervised Learning
4. **Two step convolutional neural network for automatic glottis segmentation in stroboscopic videos** [Link] 2020
   Varun Belagali, A. Rao, P. Gopikishore, R. Krishnamurthy, P. Ghosh. **Biomedical Optics Express**

Experience

Graduate Researcher | Computer Vision Lab, Stony Brook University, NY, USA Sep 2022 – Present

- Histopathology slide-level pretraining with knowledge distillation from multiple patch encoders (ongoing).
- Self-supervised learning with synthetic data. Gen-SIS: view-invariant SSL with Dino, I-BOT, SimCLR.
  CDG-MAE: cross-view SSL with Masked Autoencoders.
- HyperMAE: Implicit neural representation based decoder architecture to efficiently train Masked Autoencoders.

Research Intern | Dolby Laboratories, Sunnyvale, USA June 2025 – Aug 2025
Post-training Video-JEPA models for egocentric action anticipation.

Research Associate | Indian Institute of Science, Bangalore, India Oct 2021 – July 2022
Deep Learning for glottis segmentation in stroboscopic videos. Advised by Prof. Prasanta Kumar Ghosh.

Software Engineer | Citrix, Bangalore, India July 2020 – Sep 2021
Cloud Engineering: Development of traffic manager software to handle regional outages. Lead of cloud cost optimization.

Technical Skills

Technologies: Computer Vision, Deep Learning, Self-Supervised learning, Large-scale Distributed Pre-training.
Languages & Frameworks: Python, PyTorch, Latex
Cloud Engineering: Azure, Jenkins, Splunk, New Relic

Research Talks

Self-supervised Learning with Synthetic Data. **GenBio AI**, Palo Alto, USA (Sep 2025)

Learning Correspondences from Diffusion Generated Views. **DATAIA Talk**, CentraleSupélec, Paris, France (May 2025)

Generative Self-augmentation Improves Self-supervised Learning. **MICS Lab**, CentraleSupélec, Paris, France (March 2025); **IMAGINE Lab**, École des Ponts, Paris, France (April 2025)

Fellowships

SBU CS Chairman Fellowship (2024 - 2026), DATAIA Fellowship (2025)